

Kanchan Basnet

Kathmandu, Nepal | kanchanbasnet.dev@gmail.com | +977 9705850328 | kanchanbasnet.com.np
linkedin.com/in/kanchanbasnet01 | github.com/Kanchanbasnet

Summary

Backend Developer with **2.3 years** of industry experience specializing in **microservices, distributed systems, and AI-powered backend systems**. Skilled in Node.js, TypeScript, and PostgreSQL, with hands-on experience in LLM integration, RAG pipelines, and agentic AI systems. Passionate about scalable backend design and applying techniques to solve real-world problems.

Education

Bachelor of Science in Computer Science and Information Technology 2019 – 2024
Tribhuwan University, Padmakanya Multiple Campus | Kathmandu, Nepal
Percentage: 81.52

Experience

Software Engineer, Amnil Technologies Pvt. Ltd, Nepal Dec 2023 – Feb 2026

• Generative AI – Retrieval Augmented Generation

A real-time AI-powered chat platform that enables users to upload documents and receive context-aware answers based on their content.

Technologies Used: Node.js, Express.js, LoopBack, PostgreSQL, OpenAI API, LangChain.js, ChromaDB, MongoDB, Redis, BullMQ, Docker

- **Implemented context-aware conversations** using LangChain.js, enabling the chatbot to understand follow-up queries and maintain user context across sessions.
- **Optimized large-scale web crawling using BullMQ queues**, increasing ingestion throughput and preventing data mix-ups during concurrent processing.
- **Developed voice-based query and response system** using speech-to-text and TTS pipelines, allowing fully audio-driven chatbot interactions.
- **Integrated advanced document-processing APIs** (GraphRAG + Preeti font parser) to support complex document structures in Nepali and English.

• NCELL and Machhapuchhre Bank Customer Facing Chatbots

Intent-based NLP chatbots enabling customers to access information and services in real time via conversational interfaces.

Technologies Used: Botpress, Node.js, TypeScript, Express.js, Socket.io, React, SQLite, PostgreSQL, Redis, Docker

- **Built an analytics dashboard** tracking Daily Active User, user retention, top intents, and FAQ metrics, improving chatbot monitoring and business insights.
- **Integrated Ncell's customer APIs** to deliver personalized responses based on user profile, improving accuracy and user satisfaction.
- **Implemented a conversational interface** for MBL with a focus on an intuitive, user-friendly chat experience to improve customer engagement and ease of interaction.

• AI Growth Coach

An internal generative AI chatbot designed to assist users with document-based and web-powered querying.

Technologies Used: Node.js, Express.js, PostgreSQL, React.js

- **Developed a “What’s New” feature**, enabling admins to add descriptions, images, and videos for newly introduced application features via the CMS, which were dynamically displayed on the client side.
- **Implemented a split-screen view for sources**, enabling users to view supporting content alongside their chatbot conversation for better context and usability.

- **Refactored** backend codebase and database structure, removing deprecated modules and improving maintainability and system performance.
- **Wrote unit and integration tests** using Jest across backend services, improving reliability and reducing regressions in production

NodeJS Intern, Amnil Technologies Pvt. Ltd, Nepal

Sept 2023 – Dec 2023

- Developed and maintained RESTful APIs using **Node.js** and **Express.js**, improving backend response efficiency
- Integrated **MongoDB** / **PostgreSQL** databases and wrote optimized queries for data retrieval and storage
- Implemented **JWT-based authentication** and role-based access control (RBAC) for secure user management
- Collaborated with frontend developers to connect APIs and resolve integration issues in an **Agile** environment
- Wrote unit and integration tests using **Jest**, increasing code reliability and reducing production bugs

Technologies

Programming Languages: C, JavaScript, TypeScript

Libraries/Frameworks: React.js, Express.js, LangChain, GraphQL

Databases: MySQL, PostgreSQL, SQLite, Redis, ChromaDB

Tools/Technologies/Platforms: Docker, Git, GitHub, GitLab, Jenkins, GitHub Actions, Jest, Portainer, REST API

Projects

Multi Modal AI Chatbot

[View Github](#)

Technologies Used: Node.js, TypeScript, OpenAI, PostgreSQL, Docker, Redis

- Built a production-ready AI chatbot backend in a Turborepo monorepo with an agentic pipeline using OpenAI function calling for multi-step task execution.
- Features async job processing via a background worker, Google OAuth, PostgreSQL + Prisma, and full CI/CD via GitHub Actions deployed on Railway.

Activities

- **Vice President, PK IT CLUB (2023–2024):** Led technical events, workshops, and delivered a Git/GitHub training session for new developers.
- **Digital Marketing Head, PK IT CLUB (2022–2023):** Managed campaigns and content strategy to increase the club's online reach and engagement.
- **College Representative, CSIT Association Of Nepal (2022–2023):** Coordinated events, workshops, and communication between students and the CSIT Association.

References

Available upon request.